Filipi N. Silva

Researcher • Data scientist • Software developer

Ph.D. in Computational Physics
Research Scientist at Indiana University Network Science Institute



Hi!

I'm Filipi Nascimento Silva, Ph.D. in computational physics, currently working as a research scientist at the Indiana University Network Science Institute (IUNI).

Research

My interests include developing and implementing new techniques for analyzing, modeling and understanding real-world systems through complex networks, machine learning and data visualization.

Experience

I've published papers and developed software on many scientific fields, including bioinformatics, text analysis, scientometry, information science, and urban networks. I'm also the creator of the Networks 3D and Helios-Web visualization tools.

Future

I'm looking for opportunities around the world to continue to develop my skills and share my experience while performing research on interdisciplinary topics or bringing amazing products to life. I'm interesting in careers at the university, industry, or a combination of both.

Specs.

FULL NAME	Filipi Nascimento Silva
CONTACT INFORMATION	Indiana University Network Science Institute (IUNI) Bloomington, IN 47408, USA +1 (812) 855-6428 filipinascimento.github.io filipinascimento@gmail.com
LANGUAGES	Portuguese (native) and English
CITIZENSHIP	Brazilian
EDUCATION	Ph.D. in Physics, *Speciality: Computational Physics* 2011 – 2015 São Carlos Institute of Physics (University of São Paulo) MSc in Physics, *Speciality: Computational Physics* 2007 – 2009 São Carlos Institute of Physics (University of São Paulo) B.S in Physics, *Speciality: Computational Physics* 2003 – 2006 São Carlos Institute of Physics (University of São Paulo)
LINKEDIN	http://linkedin.com/in/filipinascimento
RESEARCHGATE	https://www.researchgate.net/profile/Filipi_Silva2
GOOGLE SCHOLAR	https://scholar.google.com.br/citations?user=fhWJEysAAAAJ
METRICS	48 publications in indexed journals 2 book chapters 471 citations (Web of Science) 499 citations (SCOPUS) 939 citations (Google Scholar)

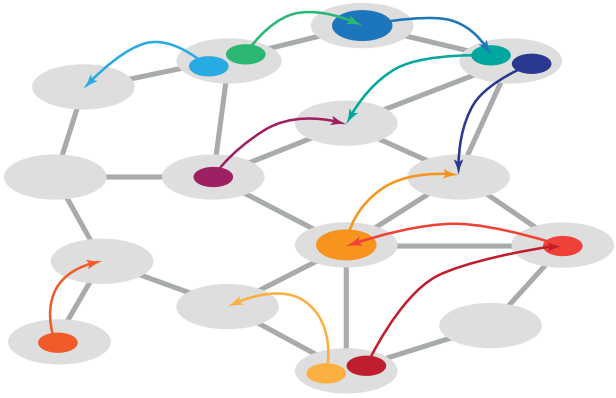
Experience



Research Scientist

IUNI - Indiana University (IU)
2019 - now

Currently studying the dynamics of science by tracking how scientific fields evolve over time.
Also working on other projects related to building networks from similarity data, such as networks of politicians connected by vote correlation. Also developing a new set of interactive web-based network visualization tools for researchers named Helios-Web.



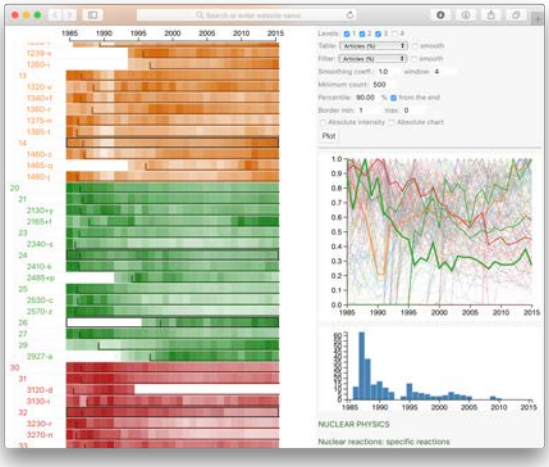
Postdoctoral Fellow

IFSC[1] - University of São Paulo (USP)
2015 - 2018

Studied dynamics occurring or driving the evolution of *complex networks* along time for a diverse range of datasets, such as texts, cities, financial market and biological systems. Also developing new web-based tools and *libraries* for *network analysis* and *visualization*, as well as new *theoretical approaches* in complex networks. Was mentor of student G. S. Domingues for the scientific initiation project: *Topological characterization of world cities,* which resulted in one publication.
Advisor: Luciano da F. Costa
(grants by FAPESP)

1 - São Carlos Institute of Physics

Experience (cont)



Visiting Researcher

SICE[1] - Indiana University
2017 - 2018

Currently studying the evolution of scientific fields along time based on information coming from content (such as texts), citation networks and popularity metrics. Our main goal is to predict the emergence of new scientific fields and understand its dynamics. For that, we developed a web application to explore popularity time-series obtained for scientific fields and properties from citation networks.

Supervision: Filippo Menczer
(grants by FAPESP)



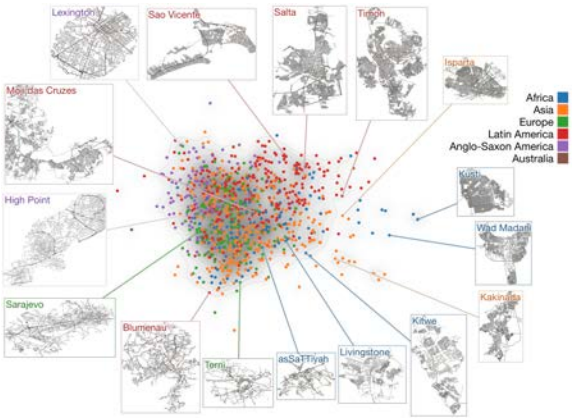
iOS and Web Developer

Freelancer
2009-2010 and 2019

Acting as freelancer consultant developer of solutions for *iOS*. Started developing an *entertainment* and *visualization library* for iOS devices written in *Objective-C* and using *OpenGL ES*. Currently developing an iOS game called *Gridland*. Notable clients: Paperless[2], Gtechnologies, Plus-TI, and Rádio Eldorado[3].

1 - School of Informatics, Computing and Engineering
2 - www.paperlessdobrasil.com.br
3 - www.facebook.com/radioeldorado/

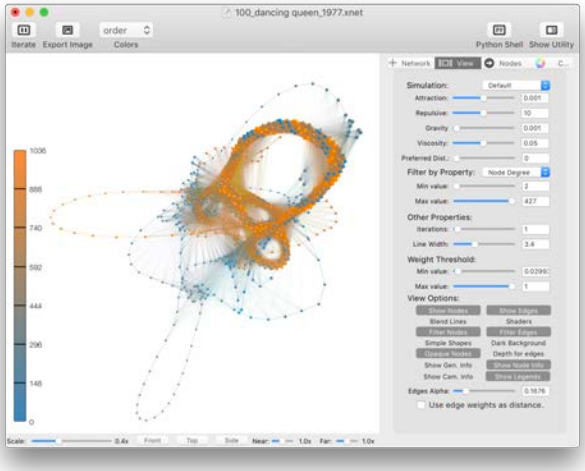
Experience (cont)



Data Scientist

Freelancer for Bike da Firma
2015

Worked as a freelancer consultant and developer of *data science* solutions for Bike da Firma[1]. Conceived and implemented *classification techniques* employed to *extract patterns* of information retrieved from mobile devices, such as *geolocation* and *accelerometer data*.



Ph.D. Thesis

IFSC - University of São Paulo (USP)
2011 - 2015

Studied the *multidimensional structure* of *complex networks* such as *fractals* and high dimensional geographic networks. We analyzed many real-world systems, such as gene to gene correlation networks, academic publication data and geographic networks. During this time I also improved and optimized the algorithms behind our network visualization tool leading to a visualizer tool for the web.
Advisor: Luciano da F. Costa
(grants by FAPESP)
Among the three awarded theses in exact sciences of USP in 2015/2016.[2]

1 - www.bikedafirma.com/en/
2 - https://jornal.usp.br/universidade/usp-anuncia-vencedores-do-premio-tese-destaque-2017/

Experience (cont)

Scientific initiation
IFSC - USP
2007 - 2009

Studied the multiscale structure of complex network derived from complex systems by means of hierarchical node-centered measurements. Developed a framework to obtain Hierarchical (Concentric) properties from complex networks and published an article in Journal of Statistical Physics. Supervision: Luciano da Fontoura Costa.
Advisor: Luciano da F. Costa
(grants by CAPES)

Scientific initiation
IFSC - USP
2004 - 2005

Ported RoundMidnight - EXAFS Datafiting software from Mac OS X to Windows.
Advisors: Valmor R. Mastelaro and Alain Michalowicz
(grants by CNPq)

Misc. experience

Teaching experience

Organized and conducted the IUNI Workshop on "An Introduction to Network Visualization" 2022
https://iuni.iu.edu/news/event/79

Organized and conducted the CADRE Workshop for the ISSI 2021
https://cadre.iu.edu/news-and-events/events/networks-2021-cadre-workshop-satellite-2

Assistant lecturer internship (Programa PAE) - 2014
Grants by CAPES / IFSC - University of São Paulo
Course: Mathematical and Computational Modeling
Duration: 120 hours Supervision: Prof. L. da F. Costa

Conducted the workshop: Introduction to Interactive Visualization of Data and Complex Networks
5ª Semana da Física / IFSC - University of São Paulo

Mentorship and supervising experience

Google Summer of Code 2021 and 2022

Mentor of Bruno Messias (2021) and Mohamed Agour (2022) for the FURY/Helios project (under the Python Software Foundation)

https://fury.gl/latest/posts/2021/2021-08-23-gsoc-devmessias-final-report.html

Scientific initiation - Topological characterization of world cities. Student: Guilherme S. Domingues 2017
Grants by CAPES / IFSC - University of São Paulo

Reviewer experience

Physics Letters A, Complexity, IEEE Transactions on Network Science and Engineering, PeerJ, IEEE DSAA, Entropy, Journal of Informetrics, Scientometrics, PLOS ONE, Nature Human Behaviour.

Selected publications

Börner, K., Silva, F. N., & Milojević, S. Visualizing big science projects. *Nature Reviews Physics*, 1-9, 2021. doi: 10.1038/s42254-021-00374-7

Comin, C.H.; Peron, T.K.DM.; Silva, F. N.; Amancio, D. R.; Rodrigues, F.A. and Costa, L. da F. Complex systems: features, similarity and connectivity. Physics Reports, 2020. doi: 10.1016/j.physrep.2020.03.002

Bando, S.Y., Iamashita, P., Silva, F.N., Costa, L.D.F., Abe, C.M., Bertonha, F.B., Guth, B.E., Fujita, A. and Moreira-Filho, C.A.
Dynamic gene network analysis of Caco-2 cell response to Shiga toxin-producing Escherichia coli-associated hemolytic–uremic syndrome. *Microorganisms,* 7(7), p.195, 2019. doi: 10.3390/microorganisms7070195

Silva, F. N.; Amancio, D. R.; Bardosova, M.; Oliveira Jr., O. N.; Costa, L. da F. Using network science and text analytics to produce surveys in a scientific topic. Journal of Informetrics, v. 10, n. 2, p. 487 – 502, 2016 doi: 10.1016/j.joi.2016.03.008

Silva, F.N.; Comin, C.H.; Peron, T.K.DM. ; Rodrigues, F.A.; Ye, C.; Wilson, R.C.; Hancock, E.R. and Costa, L. da F. Concentric network symmetry. Information Sciences, Volume 333, p. 61-80, 2015. doi: 10.1016/j.ins.2015.11.014

Moreira-Filho, C. A.; Bando, S. Y.; Bertonha, F. B.; Silva, F. N.; Costa, L. da F.; Ferreira, L.; Furlanetto, G.; Chacur, P.; Zerbini, M. C.; Carneiro-Sampaio, M. Modular Transcriptional Repertoire and MicroRNA target Analyses Characterize Genomic Dysregulation in the Thymus of Down Syndrome Infants. Oncotarget, v. 7, n. 11, p. 7497 – 7533, 2015. doi: 10.18632/oncotarget.7120

The complete list of my publications can be found in:
https://filipinascimento.github.io/publications/
https://scholar.google.com.br/citations?user=fhWJEysAAAAJ

Pre-prints and recent manuscripts can be found in arXiv:
http://goo.gl/gl5plq

Software (cont)

Helios

2020 - now



Currently, no flexible and fast tool implementing both graph layout and network rendering is available for researchers to explore and visualize large networks. The purpose of this project is to fill this gap by developing an advanced interactive visualization tool for networks based on the concepts employed on my previous works.

The project was separated into three main components: Helios Core, Helios Fury, and Helios Web.

Helios Core will implemented a set of fast and scalable layout techniques for network visualization using the C language. This includes force-directed methods and node embedding techniques. The core will serve as the basis to build the other components and will be available as a Python package or compilable to WebAssembly.

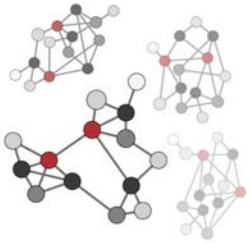
Helios Fury is a version of the rendering engine being written on top of the Fury[1] framework for Python.

Example visualizations:
https://tinyurl.com/netvizplaylist

A preliminary versions of Helios are available at:
http://github.com/filipinascimento/helios-web (helios-web)
https://heliosnetwork.io (helios-core)

Network neuroscience apps for brainlife

2020 - 2021



While network neuroscience has become popular among researchers over the past few years, the brainlife platform lacks network analysis pipelines. This subproject aims to make available network analysis and visualization tools on the brainlife framework.

Five brainlife apps were developed and can be found in:

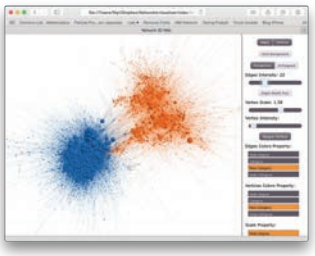
https://brainlife.io/apps#network

1 - https://fury.gl

Software (cont)

Network visualization tools for the web

2013 - 2019



To improve the usefulness of our visualization tools we started porting them to a new web-based environment. More specifically, we are reimplementing the visualizer using web technologies, such as WebGL and HTML5. Currently it can display graphically astounding complex networks visualizations on any recent major browser.

Interdisciplinary map of science obtained from a journals network (video).
http://youtu.be/ipjTMy-RrvU

Gene expression map for people suffering from febrile epilepsy (video).
http://youtu.be/Trb67DoLf5U

The software source code and demonstration can be found in:
https://filipinascimento.github.io/software/networksweb/

Networks 3D

2009 - 2019



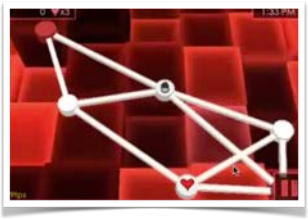
Networks 3D is a software under development, built to construct and interact with graphically appealing visualizations of large complex networks. Its main objective is to provide means for researchers and specialists to see and get a visual overview of the data they are working with. The tool is based on a optimized force-directed algorithm, which is used to project the networks to a 2D or 3D space, presenting results generated by the graphical processing unit in real-time.

More info at the software webpage:
https://filipinascimento.github.io/networks3d/

Other projects

Gridland

2011 - 2015

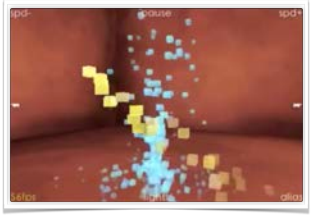


Gridland is a game being developed by me and a friend, to be distributed initially on the App Store. It is a implementation of the famous Shannon switching machine where two players compete by securing or destroying links in a graph. We developed all aspects of the game, including the artificial intelligence with 3 levels of difficulty and a 3D graphical interface. It was written entirely in C and Objective-C.

See a video of an early state version in:
http://youtu.be/-ivf89Oh0VU

HighSpace

2010 - 2011



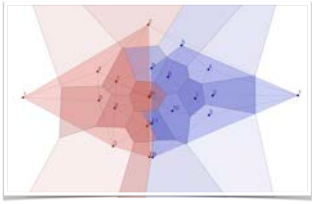
We developed a 3D scene engine on top of the Cocoa Touch frameworks and OpenGL ES. The capabilities of the engine include animated GUIs, 3D object loader, geometry representation, 3D particles engine, fast text rendering, dynamic texture atlas, keyframe animations, camera control, fast sprite rendering, etc.

Some features of the engine are shown in this video playlist:

http://www.youtube.com/playlist?list=PLFimqafBV-I5JvdrkczdCf73SKpBVN5fF
or http://goo.gl/8Rmt4

Other projects (cont)

Other assignments



Developed interactive visualizations of soccer games for data from Prof. Ricardo M. L. de Barros (UNICAMP)

http://cyvision.ifsc.usp.br/futebolTests/voronoi40000.html
(the download of the data may take several minutes)
(currently disabled)

Also developed software to obtain network metrics:
https://github.com/filipinascimento/CVAccessibility
https://github.com/filipinascimento/CVSymmetry

Developed the facebook apps for two major radio stations of São Paulo/Brasil (Rádio Eldorado and Rádio Estadão)

Created the website for the Cybernetic Vision Research Group (2012).
http://cyvision.ifsc.usp.br/
(currently disabled)

Developed a software to obtain concentric measurements of complex networks (2008).
http://cyvision.ifsc.usp.br/concentric/
(currently disabled)

Skill set summary

- Can read and write software in many programming languages such as: C, C++, Java, Objective-C, Python and Javascript. In particular, modern C language programing (C99 and C11) is my speciality. I also have some experience with the development of frameworks and libraries (see HighSpace and CVAccessibility/CVSymmetry and Helios-Web). I'm also very flexible in learning and working with another programming language.

- Experience with code analysis and debugging. Including user-level knowledge of memory management techniques, such as checking for leaks, advanced allocations and deallocation patterns, reference counting, garbage collector, retain/release cycles. Also have some experience with code profiling and optimization based on time or memory.

- Can work with parallel and distributed computing on multiple machines or CPUs, encompassing libraries such as OpenMP and OpenMPI. Also have notions of MapReduce approach and GPU-based programming. (See CVAccessibility/CVSymmetry software).

- Some knowledge of GUI development and advanced knowledge of computer graphics. (see Gridland and Networks 3D). This includes shaders and advanced rendering techniques such as impostor spheres and SDF.

- Experienced with machine learning methods, including supervised and unsupervised embedding and classification techniques, such as PCA, LDA, K_{nn}, SVM. I also have some experience using *2vec approaches to find good embeddings for data. (See papers "A pattern recognition approach to complex networks" and "Concentric network symmetry grasps authors' styles in word adjacency networks").

- Scientific Background:
  - General and advanced physics knowledge.
  - Advanced computational 3D geometry skills.
  - Advanced data analysis and statistics knowledge.
  - Can develop projects and content in a team of researchers.
  - Great skills and innovative mind to tackle new problems.

Skill set summary (cont)

- Experience with natural language processing. This includes techniques to pre-process textual data, such as lemmatization, tokenization, tf-idf and embedding. Also have experience with the proper analysis of text, including topic modeling, semantic similarity and feature extraction. (See papers "Concentric network symmetry grasps authors' styles in word adjacency networks", "Mesoscopic representation of texts as complex networks" and "Using network science and text analytics to produce surveys in a scientific topic").

- Advanced knowledge of network science and in representing arbitrary datasets as complex networks. (See experience and education sections).

- Skills to construct interactive complex visualizations of arbitrary data. (See soccer games visualization, network visualization tools in WebGL and my research papers).

- Experience with the following tools and software:
  - iOS SDK.
  - Cocoa/CocoaTouch and Objective-C libraries, including UIKit and AppKit.
  - OpenGL 2.x, OpenGL ES 1.x/2.0 and OpenGL 3.2
  - Xcode.
  - LLVM/Clang compiler, and GCC compiler.
  - Grand Central Dispatch, multi-threaded programming and blocks.

- Web developer with some experience in core technologies, such as, HTML5, CSS3, XML, JSON, WebGL, PHP, d3.js, Node.js and MongoDB. Nodejs npm package environment and deployment of ES6 web apps via snowpack and webpack.

- Remote versioning systems, such as GIT.

Most recent activities

FUNDED PROJECTS

20% FTE (AFOSR/FA9550-19-1-0391) for contributing to the science genome project (2020 - now)

OPEN SOURCE SOFTWARE

Helios Web
Implements visualization of large (up to million nodes) networks on the Web
https://github.com/filipinascimento/helios-web

Tools to process large scholarly datasets (MAG, WoS and OpenAlex)
https://github.com/filipinascimento/openalex-raw
https://github.com/filipinascimento/MAGProcessing
https://github.com/filipinascimento/WOS

Helios Core
Core component of the Helios project. Implements a parallelized force-directed layout algorithm.
https://github.com/fury-gl/helios

Robustness Modularity
Implements the R-modularity to compare detected community structure robustness across networks.
https://github.com/filipinascimento/RModularity

CVRandomWalks
Fast implementation of random walks to be used as input for node embedding methods.
https://github.com/filipinascimento/CVRandomWalks

Brainlife Network Neuroscience Apps
Basic set of apps to process and analyze networks derived from brain data.
https://doi.org/10.25663/brainlife.app.289
https://doi.org/10.25663/brainlife.app.277
https://doi.org/10.25663/brainlife.app.290
https://doi.org/10.25663/brainlife.app.321
https://doi.org/10.25663/brainlife.app.306

Most recent activities

Workshops and conferences

Presented the works *Robustness of citation networks retrieved from queries*, and *A phonetic network representation for song lyrics* at Networks 2021.

Designed and presented part of the tutorial for the *CADRE* project at the ISSI 2021 and Networks 2021 satellite.

Designed and presented part of the tutorial for the *CADRE* project at the ISSI 2019 conference.

Presented *Modeling bursts in the evolution of author citations* and *The Brazilian Chamber of Deputies as a complex network* at IC2S2 2020.

Presented works *Representing and visualizing the narrative of texts as complex networks*, *The Brazilian Chamber of Deputies as a complex network*, and *Modeling bursts in the evolution of author citations*

Grant proposal writing

Department of Defense - *Interactive exploration of dynamic complex networks at extreme scale* (pending - 2022) (role: PI)

NIH - *Integrative Study of Local Microbiome Function in Chronic Rhinosinusitis* (pending - 2022) (role: co-PI)

NSF - *Frameworks: Advancing scientific visualization for dynamic networks and time evolving data* (to be submitted - 2022) (role: co-PI)

Facebook - *Inequality in Collaboration and Mobility Dynamics of Researchers in the U.S.* Submitted to Facebook call: *Economic Opportunity and Digital Platforms request for proposals* (not funded - 2019)

Most recent publications (submitted or published)

Accessibility and Trajectory-Based Text Characterization

Bárbara C. e Souza, Filipi N Silva, Henrique F de Arruda, Luciano da F Costa, Diego R Amancio
arXiv preprint arXiv:2201.06665, 2022
https://arxiv.org/abs/2201.06665

Robustness modularity in complex networks

Filipi N Silva, Aiiad Albeshri, Vijey Thayananthan, Wadee Alhalabi, Santo Fortunato
Physical Review E, v. 105, n. 5, p. 054308, 2022
https://arxiv.org/abs/2110.02297
DOI: 10.1103/PhysRevE.105.054308

Classification of network topology and dynamics via sequence characterization

Lucas Guerreiro, Filipi N Silva, Diego R Amancio
arXiv preprint arXiv:2206.15190, 2022
https://arxiv.org/abs/2206.15190

Finding contrasting patterns in rhythmic properties between prose and poetry

Henrique Ferraz de Arruda, Sandro Martinelli Reia, Filipi Nascimento Silva, Diego Raphael Amancio, Luciano da Fontoura Costa
Physica A: Statistical Mechanics and its Applications, v. 598, p. 127387, 2022
https://arxiv.org/abs/2107.08512
DOI: j.physa.2022.127387

Detecting climate teleconnections with Granger causality

Filipi N Silva, Didier A Vega-Oliveros, Xiaoran Yan, Alessandro Flammini, Filippo Menczer, Filippo Radicchi, Ben Kravitz, Santo Fortunato
Geophysical Research Letters, v. 48, n. 18, p. e2021GL094707, 2021
https://arxiv.org/abs/2012.03848
DOI: 10.1029/2021GL094707

A comparative analysis of knowledge acquisition performance in complex networks

Lucas Guerreiro, Filipi N Silva, Diego R Amancio
Information Sciences, v. 555, p. 46--57, 2021
https://arxiv.org/abs/2007.12028
DOI: 10.1016/j.ins.2020.12.060

Most recent publications (cont.)

Visualizing big science projects

Katy Börner, Filipi Nascimento Silva, Staša Milojević
Nature Reviews Physics, v. 3, n. 11, p. 753--761, 2021
https://rdcu.be/cyEG5
DOI: 10.1038/s42254-021-00374-7

Principal component analysis: A natural approach to data exploration

Felipe L Gewers, Gustavo R Ferreira, Henrique F De Arruda, Filipi N Silva, Cesar H Comin, Diego R Amancio, Luciano da F Costa
ACM Computing Surveys (CSUR), v. 54, n. 4, p. 1--34, 2021
https://arxiv.org/abs/1804.02502
DOI: 10.1145/3447755

FURY: advanced scientific visualization

Eleftherios Garyfallidis, Serge Koudoro, Javier Guaje, Marc-Alex Côté, Soham Biswas, David Reagan, Nasim Anousheh, Filipi Silva, Geoffrey Fox, Fury Contributors
Journal of Open Source Software, v. 6, n. 64, p. 3384, 2021
https://doi.org/10.21105/joss.03384
DOI: 10.21105/joss.03384

Contrarian effects and echo chamber formation in opinion dynamics

Henrique Ferraz de Arruda, Alexandre Benatti, Filipi Nascimento Silva, Cesar Henrique Comin, Luciano da Fontoura Costa
Journal of Physics: Complexity, v. 2, n. 2, p. 025010, 2021
https://arxiv.org/abs/1910.06487
DOI: 10.1088/2632-072X/abe561

Associations between author-level metrics in subsequent time periods

Ana CM Brito, Filipi N Silva, Diego R Amancio
Journal of Informetrics, v. 15, n. 4, p. 101218, 2021
https://arxiv.org/abs/2011.12479
DOI: 10.1016/j.joi.2021.101218

On the Stability of Citation Networks

Alexandre Benatti, Henrique Ferraz de Arruda, Filipi Nascimento Silva, César H Comin, Luciano da Fontoura Costa
arXiv preprint arXiv:2105.01693, 2021
https://arxiv.org/abs/2105.01693

All other publications (submitted or published)

Enriching and analyzing small citation networks: A case study on transistor's history

Alexandre Benatti, Henrique Ferraz de Arrruda, Filipi Nascimento Silva, Luciano da Fontoura Costa
Physica A: Statistical Mechanics and its Applications, v. 573, p. 125901, 2021
https://doi.org/10.1016/j.physa.2021.125901
DOI: 10.1016/j.physa.2021.125901

Classification of abrupt changes along viewing profiles of scientific articles

Ana CM Brito, Filipi N Silva, Henrique F de Arruda, Cesar H Comin, Diego R Amancio, Luciano da F Costa
Journal of Informetrics, v. 15, n. 2, p. 101158, 2021
https://arxiv.org/abs/2005.04512
DOI: 10.31222/osf.io/fdv92

Recency predicts bursts in the evolution of author citations

Filipi Nascimento Silva, Aditya Tandon, Diego Raphael Amancio, Alessandro Flammini, Filippo Menczer, Staša Milojević, Santo Fortunato
Quantitative Science Studies, v. 1, n. 3, p. 1298--1308, 2020
https://arxiv.org/abs/1911.11926
DOI: 10.1162/qss_a_00070

Complex systems: features, similarity and connectivity

Cesar H Comin, Thomas Peron, Filipi N Silva, Diego R Amancio, Francisco A Rodrigues, Luciano da F Costa
Physics Reports, v. 861, p. 1--41, 2020
https://arxiv.org/abs/1606.05400
DOI: 10.1016/j.physrep.2020.03.002

A complex network approach to political analysis: Application to the Brazilian Chamber of Deputies

Ana Caroline Medeiros Brito, Filipi Nascimento Silva, Diego Raphael Amancio
Plos one, v. 15, n. 3, p. e0229928, 2020
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0229928
DOI: 10.1371/journal.pone.0229928

All other publications (cont.)

Opinion diversity and social bubbles in adaptive Sznajd networks

Alexandre Benatti, Henrique F de Arruda, Filipi Nascimento Silva, Cesar Henrique Comin, Luciano da Fontoura Costa
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